

Foreign Indexes trading in SGX: An empirical analysis on price discovery*

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Abstract

One of the most interesting features of the Singapore Exchange (SGX) is the trading of foreign index futures. In this paper, we examine four popular indices - Nikkiei 225 Index, MSCI Taiwan Index, CNX Nifty Index and the FTSE China A50 Index - and estimate the contribution SGX gives to the price discovery of these indices. In contrary to standard theory and evidence, we show that smaller bid-ask spread, lower minimum lots and cheaper transaction cost do not necessary improve information efficiency. These results may shed some light on the usefulness of the role of an international financial centre on the price discovery of foreign indices.

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1 Introduction

An exchange like most firms are profit maximizing firms and therefore increasing trading volume in the exchange is one of the ways an exchange is able to increase its profits. However like most firms, an exchange too faces competition for market share from other exchanges in the world, more so in today's context in which on-line trading becomes a norm linking exchanges all over the world. Methods to attract traders to traded include, but not limited to, creating innovative exchange traded products, reducing trading transaction cost, lengthening trading time to 24 hrs and increasing market depth¹. However little empirical evidence are able to conclude that these new measures will be able to increase trading volume and therefore leading the exchange to be more profitable. An exchange that managed to successfully attract foreign traders in the last few decades to trade in its trading platform is the Singapore Exchange.

One of the most interesting features of the Singapore Exchange (SGX) is the trading of foreign index futures, and being successful in doing it. An example is the trading of the Japanese Nikkei 225 index futures which flourish in SGX instead of Osaka Exchange (OSE) during the late 1990s. Many researchers easily conclude that differences in institutional characteristics, trading in the SGX is more attractive than in the OSE because of lower transaction cost and less trading restrictions, being the main reason why SGX became successful in trading Nikkei 225 futures. Fleming, Ostdiek, and Whaley (1996) proposed the trading cost hypothesis that markets with the lowest transaction cost will react more quickly to new information. This suggest that by lowering transaction cost, market can improve price efficiency therefore leading to greater price discovery. The opening of a new market trading similar products do encourage arbitrage trading which may assist in price discovery in the lagging market. On the other hand, it is hard to believe that price discovery of a country's domestic market index is coming from foreign exchange market instead of its own domestic exchange market. Unless there exist some form of barriers in the domestic exchange market that is hindering price discovery, most would expect the domestic exchange market to be the informational superior one.

¹NYSE introduce the "autoquote" system in attempt to improve liquidity.

In this paper, we hope to establish some evidence where price discovery occurs ,either domestic or foreign exchange, by investigating four different market indexes which are traded concurrently in their domestic market and SGX. The main focus of this paper is: Where price discovery occurs and why. Since SGX is a globally recognized financial exchange with regulations and standards equals or exceeds most exchanges in the world, we can examine the beneficial effects, if any, SGX can provide to the price discovery of these indexes. Besides, if SGX does provide beneficial effects to price discovery, it is also interesting to investigate factors that affect the price dicoverly process. We use the four most liquid index futures traded on SGX²: The Singapore Exchanges CNX Nifty Index Futures, FTSE China A50 Index Futures, Nikkei 225 Index Futures, MSCI Taiwan Index Futures. In order to study the price discovery of these indices, the corresponding equity-index futures are also included respectively: CNX Nifty Index listed on the National Stock Exchange of India (NSE), China Shanghai Nifty Index listed on China Financial Future Exchange (CFFEX), Nikkei 225 Index Futures listed on the Osaka Stock Exchange (OSE), and Taiwan Stock Index listed on the Taiwan Futures Exchange (TAIFEX). These futures share the same or similar underlying as those futures in SGX. These indexes allow us to compare the effects of having a well-establish exchange trading an emerging market’s index, versus a developed market’s index. In addition, this paper may shed some light on the usefulness of the role of an international financial center on price discovery of foreign indexes. Other than the exception of Japan’s Nikkei 225, the underlying of Taiwan’s Index futures traded in SGX is the MSCI Taiwan Index instead of the TAIEX Index, and the underlying of China Index traded in SGX is the FTSE China A50 Index instead of China’s CSI 300 index. As for the India’s the CNX Nifty Index, the futures trading on SGX are denominated in US dollars whereas in India, the futures are denominated in Indian Rupees. These differences are time varying and therefore controlling for these differences are important.

The main methodology we used in this paper is the information share method by Hasbrouck (1995). We find that the information share of CNX Nifty Index Futures in SGX accounts for about 77.0% and is consistently larger than that in NSE. While the information shares of Nikkei 225 Index Futures in SGX accounts for about 75.3%. On the other hand,the information shares of

²Total value of offshore futures traded in 2013 : China US\$170 billion, India US\$190 billion, Japan US\$2.8 trillion, Taiwan US\$530 billion. Detailed information can be found on <http://www.sgx.com/wps/portal/sgxweb/home/products/derivatives/overview>.

MSCI Taiwan Index Futures and FTSE China A50 Index Futures in SGX are around 27.5% and 25.1% respectively, which are all lower than the information shares of their domestic index futures. This suggest some evidence that the price discovery of local spot indices can actually happen in a foreign exchange thousands of miles away. This may come as a surprise to some as this suggest that foreign investors are more informed than the local investors or maybe the local investors are actually trading in the foreign exchange instead of its local exchange.

This paper is the first to investigate the intra-day price discovery across foreign index futures. Unlike equity which requires the access to the depository during settlements, futures are purely cash settled. Therefore trading foreign indexes through futures trading do not require any regulatory agreement from the domestic country. This paper also contribute to the price discovery literature by comparing the beneficial effect on developing foreign index futures. Investors can benefit from trading in both developed and well reputed exchanges like SGX and other developing foreign exchanges.

The rest of the paper is organized as follows. Section 2 introduce the relevant literature. Section 3 present the microstructure detail of SGX. Section 4 describe the exchange differences and the data. Section 5 explain briefly Hasbrouck (1995) information share approach. Section 6 show the empirical results on determinants affecting the informational share ratio. Section 7 investigate the cross-markets efficiency by creating trading portfolio and conclusion can be found in section 8.

2 Literature Review

Base on the efficient markets hypothesis (see Fama (1970)), all security prices in the market adjust quickly towards their efficient price, eliminating any available arbitrage profits. However in practice we do not know how this process is actually being carried out. Some suggested the existence of arbitrageurs that profit from price inefficiency which in return provide an upward or downward pressure on prices to reverse towards the efficient price. One main reason why this economic question is still unanswered is because the efficient price is not observable, but instead, we observed the traded prices, bid prices and ask prices. The law of one price do not necessary hold in reality. In

this section, we shall review some papers on the price discovery process by the futures contracts traded on different markets with similar underlyings.

2.1 Does Futures lead Spot?

The literature on whether futures prices help the price discovery process of its underlying (the spot price), gains lots of attention during the late 1990s. Using the S&P500 stock index and the futures on the stock index, Dwyer, Locke, and Yu (1996), Fleming et al. (1996), and Martens, Kofman, and Vorst (1998) showed that the futures price changes lead the spot market price change by 5 to 45 minutes. In contrast, the evidence that spot market price change lead futures price changes is weak. Tse (1999) studied the Dow Jones Industrial Average (DJIA) spot and futures price and found that most of the price discovery takes place in the futures market. In contrast, Booth, So, and Tse (1999) analysed the Germany DAX index spot, futures and options market and showed that the price discovery role is equally shared by the spot and futures market only. Although the international evidence on futures market leading the spot market is still mixed, there seemed to be a consensus that in general futures prices lead its underlying spot price.

The following are papers that investigate the price discovery process of the Japanese Nikkei 225 Index. Shyy and Shen (1997) used both daily and intra-day data to study the price transmission of Nikkei 225 futures between SIMEX in Singapore and TSE/OSE in Japan. They did not find a significant evidence of the price discovery process for both SIMEX and TSE/OSE market. Booth, Lee, and Tse (1996) used daily closing prices of Nikkei 225 Index futures from OSE, SGX and CME but found none of them can be considered the main source of information flow. Covrig, Ding, and Low (2004) examined the price discovery process using Nikkei 225 index in domestic spot market (Tokyo Stock Exchange), domestic futures market (Osaka Exchange Market) and foreign futures market (Singapore Exchange). They showed evidence that price discovery occurred in both markets and suggested that a satellite market can co-exist with another home market by providing a significant role in the price discovery process.

In relation to the price discovery process of the Taiwan Index, Chou and Lee (2002) studied

the period during tax reduction in Taiwan Futures Exchange (TAIFEX) and compared the trading costs and information transmissions between SGX and TAIFEX. They found this reduction of market friction had a great impact on the relative efficiencies of price execution of TAIFEX to SGX and the better price execution was mainly driven by the larger base of market participants and less costs of intermediation. Huang and Chou (2007) compared the difference between TAIFEX order-driven call market and SGX quote-driven continuous trading system and found the spread is minimized in TAIFEX when order imbalance is high while the spread is highest in SGX when order imbalance is high.

On the other hand little studies have been carried out on the process of prices discovery on India's Nifty Index and China's CSI 300 index. Kumar (2014) examined the impact of foreign institutional investor's investments on Nifty index futures that are both tradable on National Stock Exchange of India (NSE) and Singapore Exchange (SGX). He found that SGX futures are do not influence Nifty futures. On the other hand, Nifty futures is a significant explanatory variable in SGX futures returns. Guo, Han, Liu, and Ryu (2013) studied the intraday price discovery and volatility transmission processes between Singapore Exchange (SGX) and China Financial Futures Exchange (CFFE). They found that China's CSI 300 index futures dominate FTSE A50 index futures in SGX in both intraday price discovery and intraday volatility transmission processes.

3 SGX's Institutional Details

Singapore Exchanges derivatives market (SGX-DT) is an order driven market that uses a continuous auction system for regular intra-day trading. It provides a platform for a suite of globally tradable products, including equity index futures and options, interest rates futures and option, and dividend index futures contract. For Singapore's equity-index futures market, trading takes place during the day (T session) and during the evening (T+1 session). During the opening hours of these sessions, investors can submit orders, make amendments or cancel orders at no extra cost. SGX-DT allows investors to submit limit orders, market orders and stop orders. These orders are matched according to the price and time priority rule. SGX derivatives market face three explicit

transaction costs: exchange fees, brokerage fees and taxes. Investors pay a 0.04% of the contract value as a clearing fee and 0.0075% access fee once the order is submitted. A GST of 7% is also charged on both clearing and brokerage fees.

In addition to providing derivatives clearing and settlement, Singapore Exchange Derivatives Clearing Limited (SGX-DC) also provides a mutual offset system with Chicago Mercantile Exchange (CME). This facility allows investors to initiate positions in one exchange for allocation to the other on a real-time basis. Currently, only three index futures products are eligible for mutual offset with CME (1) Nikkei 225 Index futures (Yen denominated), (2) Nikkei 225 Index futures (USD denominated), and (3) S&P CNX Nifty Index futures.

In order to prevent excessive price volatility in the derivatives market, SGX-DT adopts price limits for the majority of its derivatives contracts. These price limits are designed to provide a cooling off period so as to restrict trading temporarily when the market is volatile. Price limits are set as a percentage of the maximum permitted movement a price can advance or decline from the previous trading days settlement price during a trading session. This specified percentage varies from contract to contract. When a price hits any of its price limits, SGX-DT will signal a cooling off period. The cooling off period is a specified duration of time where the affected contract may be traded at or within its price limits. The specified duration also varies from contract to contract. Once the cooling off period ends, normal trading resumes for the remainder of the trading day. In regards to options contracts, trading in the options contracts will be halted when their underlying futures contracts hit its price limits and enter into a cooling off period. Subsequently, normal trading for both options contracts and their underlying futures contracts will resume once the cooling off period is lifted.

4 Foreign Exchanges and Data

4.1 Foreign Exchanges

This paper use the four most liquid index futures traded on SGX: The Singapore Exchanges CNX Nifty Index Futures, FTSE China A50 Index Futures, Nikkei 225 Index Futures, MSCI Taiwan Index Futures. In order to study the price discovery of these indices, the corresponding equity-index futures are also included respectively: CNX Nifty Index listed on the National Stock Exchange of India (NSE), China Shanghai Nifty Index listed on China Financial Future Exchange (CFFEX), Nikkei 225 Index Futures listed on the Osaka Stock Exchange (OSE), and Taiwan Stock Index listed on the Taiwan Futures Exchange (TAIFEX). These futures share the same or similar underlying as those futures traded on SGX. Appendix B shows the differences between each exchanges such as the trading time, trading cost, minimum lot sizes and the multiplier. These differences affects the cost of trading in a particular exchange and the difference in cost of trading may indirectly determine the type of traders trading in the exchange. For example high-frequency speculators may want to trade in the exchange that is the cheapest to trade and the smallest size, whereas large institutional traders may not be interested in these as these factors do not affect them.

In all cases, the trading time of the four indices in their domestic exchanges is always a subset of the trading time in SGX. For the purpose of the paper, we are only interested in futures quotes and trades prices within the trading period where both the domestic and singapore exchanges open. In singapore time, this means that for the Nikkei 225 prices between 0800hrs-1415hrs are used, 0845hrs-1345hrs for the Taiwan futures index and 1145hrs-1800hrs for Nifty Index. As for the China's index futures, two sessions, 0915hrs-1130hrs and 1300hrs-1515hrs, are extracted. All futures prices not within these time interval are truncated away.

4.2 Data

The intra-day tick, time-stamped market traded, bid and ask quotes for all the index futures trading in both SGX and their home markets are obtained from Bloomberg. Our sample period span from 1st August 2014 to 31th January 2015. Intra-day foreign exchange rate data are also obtained

from Bloomberg. As the time-stamped data are accurate up the seconds, for every second, we take the lastest price to represent the price observed in that time stamped. If there are no trades or quotes in a particular second, the price in the previous time period is used instead. Trade prices, Bid and Ask prices are stored separately as individual price interval.

[Place Table 3 about here]

Table 3 reports the relative percentage of transacted index futures by the size of the lot in each exchange. For example in column 2 for the Nifty Index futures traded on SGX, 27.92% means that 27.92% of all the transacted futures contract traded on SGX have lot sizes less than 2. For comparison purposes, all lots sizes are approximately measured as a multiple of SGX lot sizes. This means that if 1 lot in Japan OSE cost twice as much as 1 lot in SGX, than we would measure the 1 lot transacted in OSE as 2 lots. Lot Ratios measures the relative cost of the smallest size lots in each exchange. For example, the cost of 1 lot in NSE is $\frac{1}{5}$ the cost of 1 lot in SGX. It cost 15.68 times more expensive to trade on CFFE than SGX, 2 times more expensive to trade on OSE than SGX and 1.76 times more expensive to trade on TFE than SGX. In conclusion, other than the Nifty Index futures on NSE, it is always cheaper to trade on SGX. Therefore what are some evidence of the consequences of having a lower minimum cost lots? From table 3, we find evidence that exchanges having a cheaper minimum lot have more smaller transactions. This may suggest that the cheaper minimum lot size is attacting certain type of global traders, traders that can trade across country borders. Whether attacting these types of traders would have a positive effect on the exchange still remains an open question. In terms of information share, due to our small international cross-sectional sample size, we are unable to test this relation. However if we were to just compare, we find no consistent relation between cheaper minimum lost size and greater information share.

[Place Table 4 about here]

From Table 4 the time-weighted spread ratio (as a percentage of futures price) is greater in SGX except for the Nikkei 225 index futures. The negative relation between cheaper minimum lot

size and smaller spread does not always hold true in our sample. The both the China and Taiwan Index futures are cheaper to trade on SGX but the spread ratio is relatively higher.

5 Price Discovery

Testing where price discovery occurs is an econometric problems and many previous papers have done do in many ways. In this paper, we shall use the information share measure constructed in Hasbrouck (1995) as our only measure. In this section, we describe briefly the information share methodology of Hasbrouck (1995) and some computational constrains used. The univariate results on information share is subsequently presented at the end of this section.

5.1 Methodology

Suppose there are n price variables related to a single security. Examples of these observable related price variables are the transaction, bid and ask prices of a traded security. In this paper, the single security of concern is an index, and the price variables related to this security are the transaction, bid and ask futures prices such that the futures underlying is the index itself. If we were to assume a fixed interest rate r over a fixed time period τ such that τ also corresponds exactly to the futures contract's time to maturity, then there exist a no arbitrage equation that relates the futures price to its underlying price: $F_\tau = S_0 e^{r\tau}$, here F_τ is the futures price with maturity τ , and S_0 is the price of the underlying at time 0. Suppose each price series is integrated of order one, $I(1)$, which implies that their price changes are covariance stationary then this means that they can be modelled using a vector moving average model (VMA):

$$\Delta p_t = \Phi(L)e_t \tag{1}$$

where e_t has $E[e_t] = 0$ and variance covariance matrix Σ . Φ is a polynomial in the lag polynomial.

Although each price is non-stationary, we know that the prices in different market do not diverge from each other significantly. Therefore we can assume that the difference between any two price is stationary, inparticular, the difference between any price variables with the first price variable is stationary. Formally this means that the prices are cointegrated of order $n - 1$ with conintegrating

matrix β :

$$\text{s.t. } \beta' = [\tau_{n-1}, -I_{n-1}]$$

$$\text{and } \beta' p_t = I(0)$$

here τ_{n-1} is a column unit vector. The requirement that $\beta' p_t$ is stationary implies that $\beta' \Phi(1) = 0$, where $\phi(1)$ is the sum of all the moving average coefficients of equation 1. We can therefore decompose the VMA model into $\delta p_t = \Phi(1)e_t + \Phi * (L)e_t$ where $\phi(1)e_t$ intuitively measures the long-run impact of a disturbance on each price variables. Given the unique structure of β , it can be shown that all the rows of $\phi(1)$ are identical which suggest a common long-run price impact on each of the price variables. Measuring the contribution from each of the price variables towards this common long-run price impact serve as a measure of information share of a market.

Since the price variables are cointegrated, there exists an error correction model (VECM) of the form:

$$\Delta p_t = \alpha(\beta' p_{t-1} - \mu) + \Gamma_1 \Delta p_{t-1} + \Gamma_2 \Delta p_{t-2} + \dots + \Gamma_K \Delta p_{t-K} + e_t \quad (2)$$

The α in equation 2 measures the speed of adjustments towards the long-run mean, μ is the long-run mean and Γ is an nxn coefficient matrix. After estimating the parameters of the VECM model in equation 2, we can then estimate the equivalent VMA model in equation 1 using the parameter of the VECM model which will be discuss in more details in the next section.

Finally, the j^{th} market information share on the single security relative to the total variance of the common random walk component can be measured as:

$$S_j = \frac{\phi_j^2 \Sigma_{jj}}{\phi \Sigma \phi'} \quad (3)$$

Here ϕ denote the common row of Φ and ϕ_j denote the j^{th} element of ϕ .

5.2 Estimation

Given observable price samples P_t , we transform it by taking natural log and define $p_t = \ln(P_t)$. Therefore the change in p_t can be the continuously compounded returns of the price samples P_t .

We estimate μ in equation 2 separately from the other parameters, and estimating it as the sample mean of $\beta'p_t$. The remaining parameters in equation 2 can then be estimated via ordinary least squares. For the purpose of this paper, we specify the VECM model to have 300 lags to account for possible autocorrelation up to 5 minutes due to uninformative trades such as trades due to inventory control purposes³.

To estimate parameters of the VMA model in equation 1, we "forecast" the VECM system subsequent to a unit perturbation. A recursive loop can then be formulated to estimate the coefficients of the VMA model. Details of this recursive formula can be found in the Appendix. We then compute $\phi(1)$ to the sum of all the moving average coefficient. In theory, $\phi(1)$ is equal to $\sum_{i=0}^{\infty} \Gamma_i$. However for the purpose of this paper, we stop the summation if:

$$||\phi_{k+1} - \phi_k||_1 < 0.0001 \quad (4)$$

ϕ_k denote $\sum_{i=0}^k \Gamma_i$ and $||M||_1$ is the matrix 1-norm. This is to ensure that convergence is reached and the system is stable. The Information Share measure in equation 3 is uniquely define if the variance-covariance matrix Σ is diagonal. This will not be the case in general, therefore we compute the average Information share by considering the Cholesky factorizations of all permutations of the disturbances.

5.3 Univariate analysis: Information Share

To compute the information Share result in table 5, both the traded price and bid-ask middle point are used as separate price variables that are cointegrated. The average information share for both price variables is later combined to compute the information share of the exchange.

[Place Table 5 about here]

From table 5, information share is greater in SGX for the Nifty index and Nikkei index. However

³Since computing technology has improved, unlike Hasbrouck (2003) we do not use any polynomial approximation of the Γ coefficients, but instead estimate all the coefficients of the Γ matrix.

the information share is lower in SGX for the China Index and Taiwan Index. All these differences are statistically significant.

6 Multivariate analysis: Factors affecting Information Share

Table 6 shows the key variable we used in the multivariate regression, and Panel A to Panel D show the detailed information for the four index futures. Both the foreign exchange rate spot price volatility and mean-adjust exchange sport price volatility are report except for the Nikkei 225 since OSE and SGX use 1000 Yen and 500 Yen as multiply respectively. Information share is the same as in univariate test and calculated from the Hasbrouck methodology. $SGX |\alpha|$ is the absolute value of the estimated speed of adjustment parameter in the VECM model and it is large for A50 and Taiwan in SGX. Besides, SGX also tend to have larger spread, higher mean-adjusted 5-minuets index futures price volatility but lower traded volume. As for the depth, SGX usually is much smaller comparing with other domestic markets except for China.

[Place Table 6 about here]

Table 7 shows the fixed effect regression of determinant of information shares. The controls are the ratio of variable in SGX divided by the corresponding in foreign derivative market. All variables are lagged by one trading day before in the consideration of contemporary information, which may incur potential endogenous problems. Mostly, we introduced lagged information share in SGX, controlling for the momentum/reversal effect of the price discovery.

The first column only control lagged information share in SGX and use fixed effect model. The coefficient of lagged information share is positive, showing a momentum effect of information share. This means that the price discovery is stably dominant in one market. Column 2 and Column 3 add additional lagged market microstructure as independent variables. We can see that the coefficients of lagged information share in SGX are much lower comparing with previous model and they are no longer significant. Most interestingly, we find a negative and marginally significant correlation of the foreign exchange spot price volatility when using SGD as denominator in Column 2 but we

do not find a significant relation of foreign exchange volatility in Column 3. This indicates that the volatility of foreign exchange may not have such a great effect on the information share. Besides, we find that the ratio of trade volume is negatively and significantly related with information shares. One percent increase in traded volume ratio leads to about 2.5% drop of information share from mean, which is quite large. This may indicate that large volume of trading is not always price efficient since mostly noisy traders may contribute to it. Finally, we see that larger depth ratio increase price efficiency in exchange with magnitudes of 2% and 1.9% in column 2 and column 3 respectively. It may be because those large traders incur with more information and help to improve the market microstructure in exchange.

[Place Table 7 about here]

7 Testing Cross-markets efficiency

The evidence that price discovery occurs in SGX for Japan's Nikkei and India's Nifty maybe a surprise from an academic view point. Information superiority from one market implies that a less informational superior market is lagging, and therefore this may lead to possible cross-market arbitrage. However in the long-run these arbitrage opportunities should not exist. Micro-structure differences such as transaction costs and bid-ask spread may lead one market to lag continuously from the other market. Literatures on limits on arbitrage may explain why such lead and lag relationships maybe exist. For examples short selling constrains or expensive trading costs maybe preventing the arbitrageurs from correcting the current price towards to the true price. Therefore understanding reasons why one market lead or lagged the other have strong economic and policy implications. In this section, we attempt to search for empirical evidence of market inefficiency by constructing trading portfolios following certain trading strategy in-line with the information share results in section ().

The trading strategy used in this paper is derived from a special-case model:

$$\begin{aligned}
P_{1,t} &= P_{1,t-1} + \epsilon_{1,t} \\
P_{2,t} &= P_{1,t-2} + \epsilon_{2,t} \\
P_{2,t} &= P_{1,t-1} - \epsilon_{1,t-1} + \epsilon_{2,t}
\end{aligned} \tag{5}$$

Since the VECM model used to derive the information share measure is general, in particular the above model 5 is a special case of the VECM, we could use information share results to construct trading strategies to exploit our main results. From model 5, we can observe that the price of a security in one market, $P_{1,t}$, follows a random walk and therefore unpredictable. However, the price of the same security in the second market is tracking the price of the first market lagged two periods. We can therefore exploit this relationship as long as we can identify which market is leading and which market is lagging. From table 7 we can infer that the daily information share of one market is very sticky and does not change through time. Therefore using this information, we shall assume throughout our sample period that the Nifty futures and Nikkei futures, SGX leads NSE and OSE respectively. Given this, our trading strategy shall be the following:

Let $P_{1,t}^{Last}$ be the last futures price of the leading market at time interval t , $P_{1,t}^{First}$ be the first futures price of the leading market at time interval t , $P_{2,t}^{Last}$ be the last futures price of the lagging market at time interval t and $P_{1,t}^{First}$ be the first futures price of the lagging market at time interval t . Information in time interval t refers to all the price information within the interval $(t-1, t]$. In this paper, trading is carried out every second from opening to closing. A buy signal is generated if the price in the leading market is higher than the price in the lagging market. This translate to $P_{1,t-1}^{Last} > P_{2,t-1}^{Last}$ and a sell signal is generated if $P_{1,t-1}^{Last} < P_{2,t-1}^{Last}$. We then compute our trading profit at the lagging market as $\pi_t = P_{2,t+1}^{First} - P_{2,t}^{First}$ if it is a buy signal and $\pi_t = P_{2,t}^{First} - P_{2,t+1}^{First}$ if it is a sell signal. This means that at time $t-1$ after we know the trading signal, we trade the first available futures price at time t and clear our position at time $t+1$ regardless whether we incur a profit or loss. To summarized,

The second portfolio we constructed in the hedged version of the trading strategy mentioned above. This means that if a long position in the lagging market is taken, a short position in the leading market is also taken, and vice versa. This strategy is popular among traders as this hedged against a sudden crash should it happen. In addition, threshold of 0%, 0.01%, 0.02%, 0.03% and 0.04% are also used for comparisons. A threshold of 0.01% means that the difference between $P_{1,t-1}^{Last}$ and $P_{2,t-1}^{Last}$ have to exceed 0.01% of $P_{1,t-1}^{Last}$ before a trading signal can be generated. Table 8 and table 9 show the summary statistics of the trading on Nifty futures and Nikkei futures respectively. In panel A, we assume that we are able to transact futures contracts at the bid-ask middle point whereas in panel B, we always buy at the ask and sell at the bid. Evidence of profits in panel B would suggest that the market is inefficient, whereas profits in panel A only suggest that one market is truly leading the other market but arbitrage profits is not possible. Figure 1 and 2 shows the cumulative daily mean rate-of-change of the Nifty futures trading portfolio, and 3 and 4 shows the cumulative daily mean rate-of-change of the Nikkei futures trading portfolio. These figures assumed that we are able to trade at the bid-ask middle point (i.e. buy and sell at the bid-ask middle point).⁴ The cumulative rate-of-change for the Nifty futures trading portfolio is increasing at a constant rate throughout the trading period from 1145hr to 1800hr. This shows that the positive profits is not due to the opening or closing trading time effects. These figures evidently show that the Nifty and Nikkei futures price trading in SGX is leading the Nifty futures price trading in NSE and the Nikkei futures price trading in OSE respectively. However if futures prices are bought at the ask and buy at the bid, the cumulative daily mean rate-of-change curve is always negative.

[Place Table 8 about here]

Table 8 shows the trading portfolio results using the Nifty Index futures trading on SGX and NSE. In panel A, the average profits (per trade) for the non-hedged portfolio is 0.005 and monotonically increased to 0.012 as the level of threshold increased. For the hedged portfolio, the average profits (per trade) is 0.012 and monotonically increased to 0.031 as the level of threshold increased. All statistics are significant at 5% level. On the other hand, if we are only able to buy at the

⁴Graphs for the cumulative daily mean rate-of-change for trades buying at the ask and selling at the bid is not shown because from Table 8 and 9 we know that the average profits are negative.

stated ask and sell at the bid, in panel B, the Average profits (per trade) is all negative. This give evidence that out trading strategy is unable to beat the bid-ask spread⁵. Similar conclusion can also be made if the same trading strategy is applied to Nikkei index futures trading on SGX and OSE in Table 9.

[Place Table 9 about here]

Evidence of negative profits in panel B does not imply that the markets are efficient. This is because the bid and ask futures prices in our sample may not be the best bid and best ask price available to trade. In addition, existence of hidden orders also known as dark pools may suggest that trading at the bid-ask middle points is also possible, but may not always possible. In conclusion, in this paper, we are not able to show evidence of efficiencies or inefficiencies of market prices.

8 Conclusion

In this paper we explore the process of price discovery of four different Indices, India's Nifty Index, China's Index, Japan's Nikkei 225 Index and Taiwan's Index. These country indices are available for trading in each of their respective domestic exchanges and in Singapore's exchange (SGX). We then investigative how the SGX contributed in the process of price discovery for each of the four indices though it index futures trading. Our results suggest that it is possible for the price discovery process to be occurring in a foreign country's exchange from the evidence of Nifty Index and Nikkei index. In addition, we construct portfolios to check if the measure is truly capturing a leading and lagging relation and show that positive profits is possible if we are able to trade at the bid-ask middle point. Although we are unable to reject the efficient market hypothesis from the negative profit evidence of our second portfolio where we have to buy at the ask and sell at the bid, we are still able to show that one market is truly leading the other. We attempt to find factors that may affect information share but at the moment most factors do not explain much.

⁵For robustness check, instead of trading every seconds, we trade every minute and similar conclusion can be drawn.

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A "Forecasting" Method

The VMA model is:

$$\Delta p_t = B_0 e_t + B_1 e_{t-1} + B_2 e_{t-2} + \dots \text{ where, } B_0 = I_n$$

and the VECM model is:

$$\Delta p_t = \alpha(\beta' p_{t-1} - \mu) + \Gamma_1 \Delta p_{t-1} + \Gamma_2 \Delta p_{t-2} + \dots + \Gamma_K \Delta p_{t-K} + e_t$$

B Exchange Differences

Table 1: Trading difference for Nikkei 225 Index Futures, MSCI Taiwan Index Futures, CNX Nifty Index Futures and FTSE China A50 Index Futures trading in SGX.

Exchange	SGX		SGX	
Underlying Stock Index	Nikkei 225 Index		MSCI Taiwan Index	
Multiplier	Y500		US\$100	
Minimum Price Fluctuation	Outright : 5 index points		0.1 index points	
	Strategy Trades: 1 index point			
Settlement Procedure	Cash Settlement		Cash Settlement	
Contract Months	6 nearest serial months		2 nearest serial months	
	20 nearest quartely months		12 nearest quartely months	
Trading Costs	Clearing Fee	0.04%	Clearing Fee	0.04%
	Trading Access Fee	0.0075%	Trading Access Fee	0.0075%
Trading Hours	T Session:		T Session:	
	Pre -Opening	07:30-07:43	Pre -Opening	08:30-08:43
	Non -Cancel Period	07:43-07:45	Non -Cancel Period	08:43-08:45
	Opening	07:45-14:25	Opening	08:45-13:45
	Pre-Closing	14:25-14:29	Pre-Closing	13:45-13:49
	Non-Cancel Period	14:29-14:30	Non-Cancel Period	13:49-13:50
	T+1 Session:		T+1 Session:	
	Pre -Opening	15:00-15:13	Pre -Opening	14:20-14:33
	Non -Cancel Period	15:13-15:15	Non -Cancel Period	14:33-14:35
	Opening	15:15-02:00	Opening	14:35-02:00
Exchange	SGX		SGX	
Underlying Stock Index	CNX Nifty Index		FTSE China A50 Index	
Multiplier	US\$2		US\$1	
Minimum Price Fluctuation	0.5 index points		5 index points	
Settlement Procedure	Cash Settlement		Cash Settlement	
Contract Months	2 nearest serial months		2 nearest serial months	
	4 nearest quartely months		4 nearest quartely months	
Trading Costs	Clearing Fee	0.04%	Clearing Fee	0.04%
	Trading Access Fee	0.0075%	Trading Access Fee	0.0075%
Trading Hours	T Session:		T Session:	
	Pre -Opening	08:45-08:58	Pre -Opening	08:45-08:58
	Non -Cancel Period	08:58-09:00	Non -Cancel Period	08:58-09:00
	Opening	09:00-18:10	Opening	09:00-15:55
	Pre-Closing	18:10-18:14	Pre-Closing	15:55-15:59
	Non-Cancel Period	18:14-18:15	Non-Cancel Period	15:59-16:00
	T+1 Session:		T+1 Session:	
	Pre -Opening	19:00-19:13	Pre -Opening	16:30-16:38
	Non -Cancel Period	19:13-19:15	Non -Cancel Period	16:38-16:40
	Opening	19:15-02:00	Opening	16:40-02:00

Table 2: Trading difference for Nikkei 225 Index Futures, MSCI Taiwan Index Futures, CNX Nifty Index Futures and FTSE China A50 Index Futures trading in their home exchanges.

Exchange	Osaka Stock Exchange		Taiwan Futures Exchange	
Underlying Stock Index	Nikkei 225 Index		TAIEX Index	
Multiplier	Y1000		NT\$200	
Minimum Price Fluctuation	0.01 index points		1 index points	
Settlement Procedure	Cash Settlement		Cash Settlement	
Contract Months	Jun and Dec: 10 nearest contract months Mar and Sep: 3 nearest contract months		2 nearest serial months 3 nearest quartely months	
Trading Costs	Clearing Fee (Proprietary)	Y20	Trasaction Fee	NT\$12
	Clearing Fee (Customer)	Y20	Clearing Fee	NT\$8
	Trading Fee (Proprietary)	Y70	Settlement Fee	NT\$8
	Trading Fee (Customer)	Y110	Futures Transaction Tax	0.0002%
Time Zone Difference	1 hour ahead		Same	
Trading Hours	Day Session		Regular Trading Days	
	Pre-Opening	08:00-09:00	Trading Hours	08:45-13:45
	Opening Auction	09:00		
	Regular Session	09:00-15:10		
	Pre-Closing	15:10-15:15		
	Closing Auction	15:15		
	Night Session			
	Pre-Opening	16:15-16:30		
	Opening Auction	16:30		
	Regular Session	16:30-02:55		
	Pre-Closing	02:55-03:00		
	Closing Auction	03:00		

Exchange	National Stock Exchange of India		China Financial Futures Exchange	
Underlying Stock Index	CNX Nifty Index		CSI 300 Index	
Multiplier	Re.1		CNY 300	
Minimum Price Fluctuation	0.5 index points		0.2 index point	
Settlement Procedure	Cash Settlement		Cash Settlement	
Contract Months	3 nearest serial months		2 nearest serial months 2 nearest quartely months	
Trading Costs	Transactions Tax(SELL only)	0.01%	Trading Fee	CNY 30
	Transaction Charges	0.00185%		
	SEBI Turnover Charges	0.0001%		
	Stamp Duty	0.002%		
Time Zone Difference	2:30 hour later		Same	
Trading Hours	Regular Trading Days		Regular Trading Days	
	Normal Market	09:15-15:30	First Session	09:15-11:30
	Setup Cutoff Time	16:15	Second Session	13:00-15:15
	Trade Modification	16:15		

Table 3: This table reports the relative percentage of transacted index futures by the size of the lot in each exchange. (i.e. 27.92 % means that 27.92% of the CNX Nifty Index futures traded on the Singapore Exchange are less than or equal to 2 lots.) All lot sizes for the foreign index futures are scaled such that they are comparable if they are traded at SGX instead. The last column, Ratios, shows the difference between 1 lot sold on SGX against 1 lot sold on the foreign exchange. (i.e. NSE Ratio of 0.21 implies that a lot sold on the NSE is 0.21 times the size of a lot sold on SGX.) SGX: Singapore Stock Exchange, NSE: National Stock Exchange of India, CFFE: China Financial Futures Exchange, OSE: Osaka Stock Exchange, TFE: Taiwan Futures Exchange.

	Lot Sizes					Lot
CNX Nifty Index	≤ 2	3 to 5	6 to 10	11 to 20	> 20	Ratios
SGX	27.92%	18.55%	15.47%	14.28%	23.79%	1.00
NSE	59.11%	20.87%	9.94%	5.79%	4.28%	0.21
A50 / CSI300	≤ 2	3 to 5	6 to 10	11 to 20	> 20	Ratios
SGX	41.73%	23.45%	17.52%	10.62%	6.69%	1.00
CFFE	0.00%	0.00%	0.00%	3.30%	96.70%	15.68
Nikkei 225 Index	≤ 2	3 to 5	6 to 10	11 to 20	> 20	Ratios
SGX	73.31%	17.28%	6.51%	1.99%	0.92%	1.00
OSE	42.28%	18.63%	17.27%	9.60%	12.21%	2.00
Taiex / MSCI Taiwan Index	≤ 2	3 to 5	6 to 10	11 to 20	> 20	Ratios
SGX	76.97%	16.77%	4.75%	0.98%	0.53%	1.00
TFE	58.06%	16.57%	15.86%	6.14%	3.38%	1.76

Table 4: This table reports the summary statistics of our sample for each of the exchanges. Panel A summarizes the Nifty Index futures traded on SGX and NSE, panel B summarizes the China's Index futures traded on SGX and CFFEC, panel C summarizes Nikkei 225 index futures traded on SGX and OSE, and panel D summarizes the Taiwan index futures trading on SGX and TAIFEX. Spread is measures as the time-weighted bid-ask spread as a ratio of the futures price. All lot sizes are converted to a relative lot size if they were to trade on SGX, using the lot ratio computed in table 3.

A)	CNX Nifty(SGX)			CNX Nifty(NSE)		
Spread (%)	0.0144			0.0081		
	Traded	Bid	Ask	Traded	Bid	Ask
Mean	43,537	100,312	100,336	68,799	72,658	74,784
Median	34,915	101,824	98,596	58,788	66,823	70,161
Std. Dev.	29,934	28,291	30,798	37,429	22,833	24,507
Std. Error	2,744	2,593	2,823	3,431	2,093	2,247
B)	FTSE China A50 Index Futures (SGX)			CSI 300 Index Futures (CFFEX)		
Spread (%)	0.0622			0.0101		
	Traded	Bid	Ask	Traded	Bid	Ask
Mean	179,800	3,093,629	3,038,831	15,254,105	12,462,827	21,304,061
Media	160,213	3,101,777	3,178,359	13,292,795	13,357,524	13,526,378
Std. Dev	104,721	1,548,610	1,460,299	7,715,988	4,812,029	96,066,204
Std. Error	9,442	139,633	131,671	695,727	433,886	8,661,998
C)	Nikkei 225 (SGX)			Nikkei 225 (OSE)		
Spread (%)	0.0316			0.0607		
	Traded	Bid	Ask	Traded	Bid	Ask
Mean	73,598	553,738	535,572	134,954	8,417,672	16,552,275
Median	68,785	518,868	502,762	127,920	7,869,585	15,024,771
Std. Dev.	38,521	203,378	198,292	70,669	3,256,482	5,955,347
Std. Error	4,419	23,329	22,746	8,106	373,544	683,125
D)	MSCI Taiwan Index Futures (SGX)			TAIEX Index Futures (TAIFEX)		
Spread (%)	0.031			0.0117		
	Traded	Bid	Ask	Traded	Bid	Ask
Mean	62,040	749,873	749,935	177,865	4,613,973	4,597,943
Median	38,448	557,481	561,590	174,615	4,446,020	4,281,071
Std. Dev.	200,114	1,874,906	1,741,393	50,593	1,326,955	1,503,804
Std. Error	18,344	171,872	159,633	4,638	121,642	137,853

Table 5: This table reports the univariate results on the daily information share ratio. The price vector used to compute the information share is traded price and bid-ask middle point. Daily information share is computed and the one sided t test is used. Superscript ***, **, and * denotes statistical significance at the 1%, 5%, and 10% levels, respectively and the t-statistics are shown in parenthesis.

	CNX Nifty(SGX)	CNX Nifty(NSE)	Difference
Mean	63.02%	36.98%	26.04%***
Std. Error	2.11%	2.11%	4.22%
Obs	119	119	119
	FTSE China A50 Index Futures (SGX)	CSI 300 Index Futures (CFFEX)	Difference
Mean	22.79%	77.21%	-54.42%***
Std. Error	1.47%	1.47%	2.94%
Obs	123	123	123
	Nikkei 225 (SGX)	Nikkei 225 (OSE)	Difference
Mean	61.19%	38.81%	22.38%***
Std. Error	1.75%	1.75%	3.50%
Obs	95	95	95
	MSCI Taiwan Index Futures (SGX)	TAIEX Index Futures (TAIFEX)	Difference
Mean	25.18%	74.82%	-49.64%***
Std. Error	1.73%	1.73%	3.46%
Obs	127	127	127

Table 6: Add caption

Panel A: CNX Nifty Index Futures(SGX) / CNX Nifty Index Futures(NSE)					
FX σ	0.0809				
FX σ^*	0.0013				
	SGX	NSE	Difference	T-statistics	
IS	0.7699	0.2301	0.540	11.237	***
$ \alpha $	0.0036	0.0056	-0.002	-4.146	***
Spread	0.0142	0.0082	0.006	20.070	***
Price σ^*	0.0021	0.0021	0.000	4.423	***
Depth	8.8469	32.665	-23.818	-23.041	***
Log(Traded Volume)	10.521	11.08	-0.559	-8.706	***
Panel B: FTSE China A50 Index Futures(SGX) / CSI 300 Index Futures(CFFE)					
FX σ	0.0036				
FX σ^*	0.0006				
	SGX	CFFEX	Difference	T-statistics	
IS	0.2746	0.7254	-0.451	-8.413	***
$ \alpha $	0.0097	0.0004	0.009	3.205	***
Spread	0.0586	0.0101	0.049	27.213	***
Price σ^*	0.0066	0.0057	0.001	4.653	***
Depth	364.88	137.7	227.188	5.075	***
Log(Traded Volume)	11.983	16.416	-4.434	-62.776	***
Panel C: Nikkei 225 Index Futures(SGX) / Nikkei 225 Index Futures (OSE)					
	SGX	OSE	Difference	T-statistics	
IS	0.7537	0.2463	0.507	12.897	***
$ \alpha $	0.0142	0.0285	-0.014	-8.280	***
Spread	0.0315	0.061	-0.030	-54.583	***
Price σ^*	0.0024	0.0025	0.000	-3.497	***
Depth	54.926	591.41	-536.481	-29.598	***
Log(Traded Volume)	11.098	11.711	-0.613	-49.855	***
Panel D: MSCI Taiwan Index Futures(SGX) / TAIEX Index Futures(TFE)					
	SGX	TAIFEX	Difference	T-statistics	
FX σ	0.0276				
FX σ^*	0.0009				
IS	0.2516	0.7484	-0.497	-10.346	***
$ \alpha $	0.0021	0.0006	0.001	10.012	***
Spread	0.0304	0.0118	0.019	17.821	***
Price σ^*	0.0023	0.002	0.000	6.068	***
Depth	82.215	302.71	-220.492	-10.623	***
Log(Traded Volume)	10.614	12.123	-1.509	-23.779	***
Panel E: Ratio					
	Obs	Mean	Std.Dev.		
Spread Ratio	417	2.7736	2.32776		
Volatility Ratio	417	1.1021	0.2377724		
Traded Volume Ratio	417	0.401	0.7171959		
Depth Ratio	417	1.2213	2.287196		

Table 7: The table examines the factors that explain the proportion of information share contributed by an exchange. The dependent variable is SGX IS. Subscript t denotes calendar day. SGX IS is the proportion of information share attributed to SGX. $\text{FX } \sigma^{SGD}$ is the mean-adjusted 5-minutes foreign exchange spot price (Foreign Currency/SGD) volatility. $\text{FX } \sigma^{USD}$ is the mean-adjusted 5-minutes foreign exchange spot price (Foreign Currency/USD) volatility. Spread Ratio is the spread of the index futures traded on SGX divided by the spread of the index futures traded on its domestic exchange. Price σ^* is the mean-adjusted 5-minutes index futures price volatility. SGX $|\alpha|$ is the absolute value of the estimated speed of adjustment parameter in the VECM model. Volume Ratio is the daily traded volume of the index futures on SGX divided by the daily traded volume of its corresponding foreign index futures. Depth Ratio is the Depth of SGX index futures divided by the Depth of its corresponding foreign index futures. Depth is measured as the average 1-second bid and ask total volume. Volume is measured as the number of lots multiply by the exchange respective lot ratio found in Table 3. Superscript ***, **, and * denotes statistical significance at the 1%, 5%, and 10% levels, respectively and the t-statistics are shown in parenthesis.

SGX IS _t	(1)	(2)	(3)
SGX IS _{t-1}	0.165** (0.051)	0.109 (0.110)	0.110 (0.102)
FX σ^{SGD}_{t-1}		-36.217* (12.207)	
FX σ^{USD}_{t-1}			21.400 (13.242)
Spread Ratio _{t-1}		-0.048 (0.027)	-0.049 (0.028)
Price σ^*_{t-1}		0.005 (0.088)	0.006 (0.083)
SGX $ \alpha _{t-1}$		0.058 (0.803)	0.016 (0.767)
Volume Ratio _{t-1}		-0.025* (0.010)	-0.025* (0.009)
Depth Ratio _{t-1}		0.020** (0.004)	0.019** (0.004)
Constant	0.634*** (0.027)	0.817** (0.200)	0.730** (0.211)
Index Fixed Effect	Yes	Yes	Yes
Day Fixed Effect	Yes	Yes	Yes
N	315	315	315
Adj. R ²	0.462	0.483	0.481

Table 8: Nifty Futures:

Trade/Sec	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	NoHedged	Threshold = 0%	NoHedged	Threshold = 0.01%	NoHedged	Threshold = 0.02%	NoHedged	Threshold = 0.03%	NoHedged	Threshold = 0.04%
	Hedged	Hedged	Hedged	Hedged	Hedged	Hedged	Hedged	Hedged	Hedged	Hedged
Panel A: Midpoint										
A.Profit (per trade)	0.005	0.012	0.007	0.014	0.008	0.017	0.009	0.022	0.012	0.031
t-stat	27.09	3.38	27.86	3.19	27.92	2.90	27.12	2.67	28.88	2.54
A.Return (per trade)	0.00007%	0.00009%	0.00008%	0.00012%	0.00009%	0.00014%	0.00011%	0.00019%	0.00015%	0.00026%
t-stat	27.13	2.33	27.88	2.23	27.92	2.08	27.06	1.96	28.79	1.89
A.Daily Profits	121.79	259.12	112.12	244.58	99.51	222.65	84.39	205.06	76.63	195.03
t-stat	10.42	3.31	11.97	3.15	14.24	2.90	13.80	2.70	10.94	2.58
A.Daily Returns	1.49%	1.57%	1.37%	1.49%	1.22%	1.35%	1.03%	1.25%	0.94%	1.21%
t-stat	10.35	3.28	11.93	3.12	14.12	2.88	13.72	2.65	10.88	2.48
Skewness	1.30	8.15	1.26	8.38	1.02	8.40	0.79	8.53	1.13	8.67
Kurtosis	4.94	73.59	5.54	76.30	4.84	75.87	3.04	78.09	3.65	80.75
Maximum Drawdown	-18.73	-32.27	-13.52	-32.27	-13.52	-32.27	-13.52	-32.27	-13.52	-32.27
Panel B: Buy Ask/Sell Bid										
A.Profit (per trade)	-0.658	-1.840	-0.664	-1.835	-0.668	-1.824	-0.668	-1.816	-0.667	-1.807
t-stat	-2033.84	-409.31	-1766.89	-312.39	-1517.72	-232.77	-1274.43	-167.42	-1045.92	-114.79
A.Return (per trade)	-0.00803%	-0.01119%	-0.00811%	-0.01114%	-0.00815%	-0.01106%	-0.00815%	-0.01099%	-0.00816%	-0.01092%
t-stat	-2034.80	-211.01	-1767.33	-160.71	-1518.24	-119.44	-1275.28	-85.71	-1047.52	-58.67
A.Daily Profits	-14775.28	-41324.43	-11402.69	-31493.52	-8582.98	-23438.55	-6195.00	-16842.82	-4257.23	-11538.39
t-stat	-78.40	-83.03	-29.56	-31.68	-17.06	-17.27	-11.90	-11.85	-9.22	-9.04
A.Daily Returns	-180.30%	-252.03%	-139.13%	-192.08%	-104.69%	-142.93%	-75.61%	-102.77%	-52.10%	-70.58%
t-stat	-76.88	-82.84	-29.43	-31.49	-17.00	-17.20	-11.83	-11.78	-9.15	-8.98
Skewness	-0.35	-0.74	0.23	0.35	-0.08	0.00	-0.54	-0.49	-1.00	-0.96
Kurtosis	3.22	3.23	2.32	2.42	1.76	1.79	1.90	1.88	2.66	2.68
Maximum Drawdown	-21.20	-42.95	-21.20	-42.95	-21.20	-42.95	-21.20	-42.95	-21.20	-42.95
Daily No. of Trades	22,456	22,456	17,165	17,165	12,849	12,849	9,275	9,275	6,387	6,387
Sample Size	119	119	119	119	119	119	119	119	119	119

Table 9: Nikkei Futures:

Trade/Sec	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Threshold = 0% NoHedged	Hedged	Threshold = 0% NoHedged	Threshold = 0.01% Hedged	Threshold = 0.02% NoHedged	Threshold = 0.02% Hedged	Threshold = 0.03% NoHedged	Threshold = 0.03% Hedged	Threshold = 0.04% NoHedged	Threshold = 0.04% Hedged
Panel A: Midpoint										
A.Profit (per trade)	0.115	0.129	0.117	0.131	0.487	0.530	0.483	0.526	0.475	0.514
t-stat	125.16	137.33	125.65	139.07	40.93	44.60	34.54	37.66	30.92	33.45
A.Return (per trade)	0.00070%	0.00040%	0.00071%	0.00040%	0.00294%	0.00160%	0.00295%	0.00160%	0.00288%	0.00156%
t-stat	125.55	137.95	126.04	139.68	40.89	44.60	34.63	37.80	30.91	33.44
A.Daily Profits	2564.53	2881.79	2564.21	2884.79	342.47	372.68	263.11	286.08	224.42	242.68
t-stat	17.13	19.80	17.14	19.80	11.59	13.05	12.20	15.04	12.58	14.20
A.Daily Returns	15.66%	8.81%	15.65%	8.82%	2.07%	1.13%	1.61%	0.87%	1.36%	0.73%
t-stat	17.55	20.39	17.56	20.40	11.89	13.56	12.30	15.31	12.95	14.80
Skewness	0.76	0.58	0.76	0.58	1.86	1.31	3.42	2.01	1.78	1.24
Kurtosis	2.82	2.61	2.84	2.61	7.80	4.62	22.03	10.47	7.50	4.78
Maximum Drawdown	-70	-70	-70	-70	-70	-70	-70	-70	-70	-70
Panel B: Buy Ask/Sell Bid										
A.Profit (per trade)	-9.73014	-14.8253	-9.72623	-14.7863	-6.04633	-11.9228	-5.0671	-10.5306	-4.42051	-9.57728
t-stat	-7556.19	-10509.62	-7462.86	-10621.69	-281.68	-480.90	-200.96	-379.33	-162.00	-334.18
A.Return (per trade)	-0.06015%	-0.04581%	-0.06014%	-0.04571%	-0.03643%	-0.03603%	-0.03113%	-0.03240%	-0.02678%	-0.02915%
t-stat	-7312.01	-9923.08	-7224.62	-9980.01	-280.61	-483.64	-200.98	-376.99	-161.61	-336.12
A.Daily Profits	-216982.74	-330605.74	-213711.68	-324895.58	-4253.05	-8386.63	-2758.32	-5732.42	-2086.53	-4520.58
t-stat	-111.73	-111.91	-109.91	-111.00	-13.11	-14.87	-16.72	-20.20	-14.62	-19.21
A.Daily Returns	-1341.30%	-1021.61%	-1321.50%	-1004.33%	-25.62%	-25.34%	-16.94%	-17.64%	-12.64%	-13.76%
t-stat	-92.71	-94.69	-89.21	-91.12	-13.68	-15.67	-16.64	-20.09	-15.19	-20.32
Skewness	5.63	6.00	5.04	5.37	-0.88	-0.89	-1.10	-1.25	-0.92	-0.90
Kurtosis	46.35	50.52	40.01	43.56	3.20	3.19	4.20	4.89	3.56	3.58
Maximum Drawdown	-70	-105	-70	-105	-70	-105	-70	-105	-70	-105
Daily No. of Trades	22,300	22,300	21,973	21,973	703	703	544	544	472	472
Sample Size	95	95	95	95	95	95	95	95	95	95

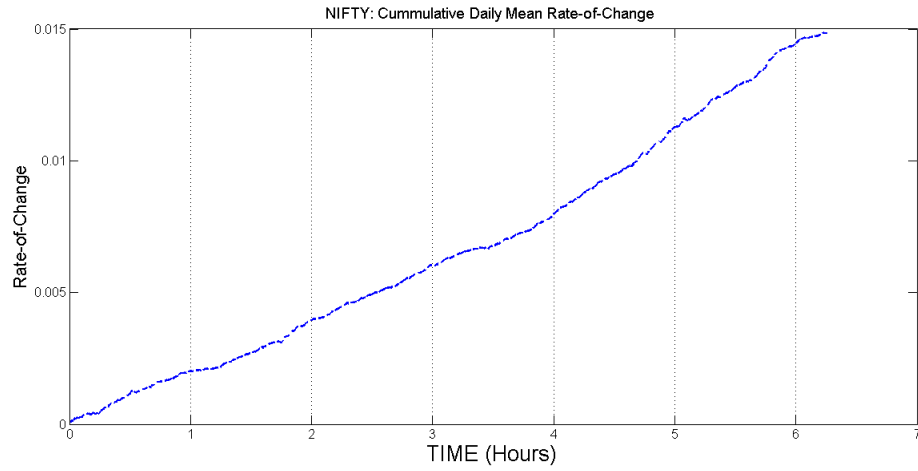


Figure 1: Nifty futures trading portfolio (un-hedged) daily cumulative mean rate-of-change.

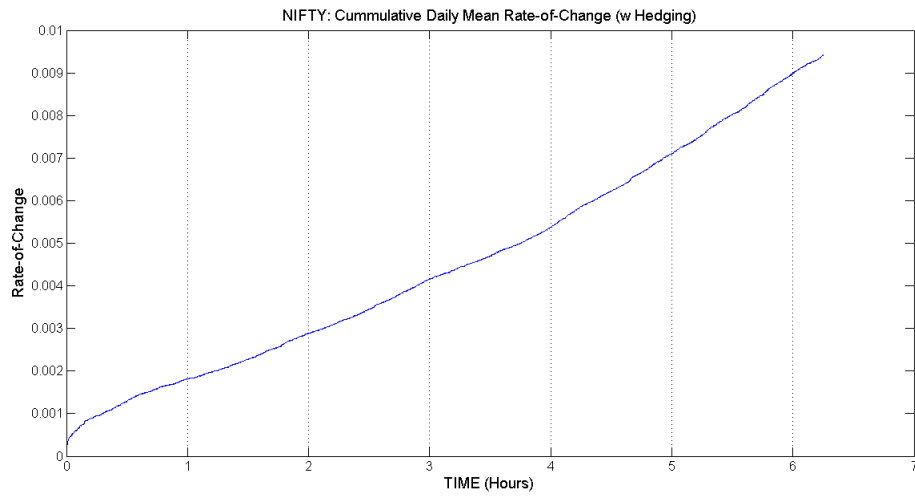


Figure 2: Nifty futures trading hedge portfolio daily cumulative mean rate-of-change.

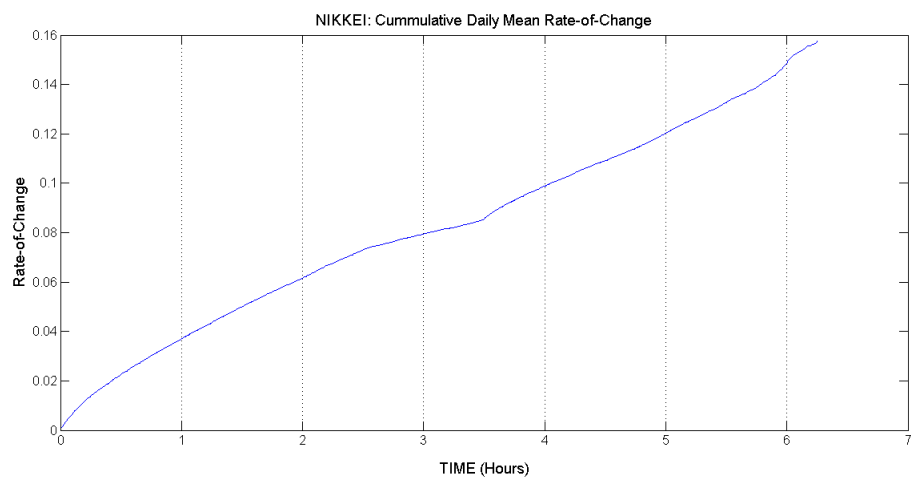


Figure 3: Nikkei futures trading portfolio (un-hedged) daily cumulative mean rate-of-change.

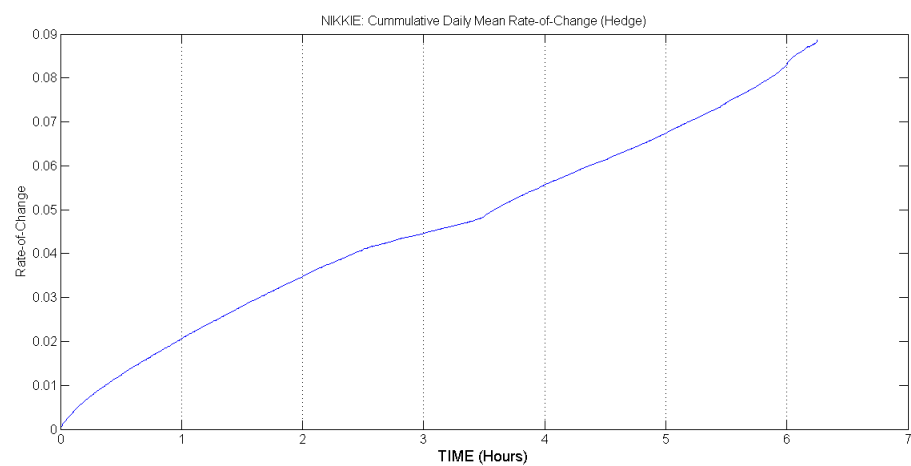


Figure 4: Nikkei futures trading hedge portfolio daily cumulative mean rate-of-change.